System of Differential Equations:

⇒ Let $y' = \frac{dy}{dx} = f_1(x, y, z)$

$$z' = \frac{dz}{dx} = f_2(x, y, z) \quad , \; y(x_0) = y \;\&\; z(x_0) = z_0$$

1> Euler's Method:

⇒ $y_{n+1} = y_n + h f_1(x_n, y_n, z_n)$

$z_{n+1} = z_n + h f_2(x_n, y_n, z_n)$; $n = 0, 1, 2, 3, \ldots$

*** choose h such that $n = 0, 1, 2 \& 3$.

2> Heun's Method:

⇒ ~~$y_{n+1} = y_n + h f_1(x_n, y$~~

$$y_{n+1} = y_n + \frac{h}{2}(m_1 + m_2)$$

$$z_{n+1} = z_n + \frac{h}{2}(K_1 + K_2)$$

where,

$m_1 = f_1(x_n, y_n, z_n)$

$K_1 = f_2(x_n, y_n, z_n)$

$m_2 = f_1(x_n + h, y_n + m_1 h, z_n + K_1 h)$

$K_2 = f_2(x_n + h, y_n + m_1 h, z_n + K_1 h)$; $n = 0, 1, 2, \ldots$

*** choose h such that $n = 0 \& 1$.

* Numericals:

Q.1> Solve $\frac{dy}{dx} = x+z$, $\frac{dz}{dx} = x-y^2$, $y(0)=2$, $z(0)=1$.

Using,

i> Euler's Method

ii> Heun's Method, Estimate y at 1 and z at 1.

Sol^n Here,

$$\frac{dy}{dx} = x+z, \quad y(0)=2$$

$$\frac{dz}{dx} = x-y^2, \quad z(0)=1$$

$\therefore f_1(x,y,z) = x+z$; $x_0 = 0$, $y_0 = 2$

$\therefore f_2(x,y,z) = x-y^2$; $\cancel{x_0} = 0$, $z_0 = 1$

1> By Euler's Method:

$$y_{n+1} = y_n + hf_1(x_n, y_n, z_n)$$

$$z_{n+1} = z_n + hf_2(x_n, y_n, z_n)$$

Take $h = 0.25$.

For $n=0$:

$y_1 = y_0 + hf_1(x_0, y_0, z_0) = 2 + 0.25(0+1) = 2.25 = y(0.25)$

$z_1 = z_0 + hf_1(x_0, y_0, z_0) = 1 + 0.25(0-4) = 0 = (0.25)$

For $n=1$:

$y_2 = y_1 + hf_1(x_1, y_1, z_1) = 2.25 + 0.25 f_1(0.25, 2.25, 0)$

$= \cancel{3+2}\ 2.3125 = y(0.5)$

$z_2 = z_1 + hf_2(x_1, y_1, z_1) = 0 + 0.25 f_2(0.25, 2.25, 0)$

$= -1.2031 = z(0.5)$

For $n=2$:

$y_3 = y_2 + h f_1(x_2, y_2, z_2) = 2.3125 + 0.25 f_1(0.5, 2.3125, -1.2031)$
$= 2.3167 = y(0.75)$

$z_3 = z_2 + h f_2(x_3, y_3, z_3) = -1.2031 + 0.25 f_2(0.5, 2.3125, -1.2031)$
$= -2.4150 = z(0.75)$

For $n=3$:

$y_4 = y_3 + h f_1(x_3, y_3, z_3) = 2.3167 + 0.25 f_1(0.75, 2.3167, -2.4150) = 1.7205 = y(1)$. Ans

$z_4 = z_3 + h f_2(x_3, y_3, z_3) = -2.4150 + 0.25 f_2(0.75, 2.3167, -2.4150) = -3.3689 = z(1)$. Ans

2) By Heun's Method:

$$y_{n+1} = y_n + \frac{h}{2}(m_1 + m_2)$$
$$z_{n+1} = z_n + \frac{h}{2}(k_1 + k_2)$$

Take $h = 0.5$

For $n=0$:

$$y_1 = y_0 + \frac{h}{2}(m_1 + m_2) \quad \text{——— (1)}$$
$$z_1 = z_0 + \frac{h}{2}(k_1 + k_2) \quad \text{——— (2)}$$

where,

$m_1 = f_1(x_0, y_0, z_0) = f_1(0, 2, 1) = 1$

$k_1 = f_2(x_0, y_0, z_0) = f_2(0, 2, 1) = -4$

$m_2 = f_1(x_0 + h, y_0 + m_1 h, z_0 + k_1 h) = f_1(0.5, 2.5, -1) = -0.5$

$k_2 = f_2(x_0 + h, y_0 + m_1 h, z_0 + k_1 h) = f_2(0.5, 2.5, -1) = -5.75$

From (1) & (2);

$y_1 = 2 + \frac{0.5}{2}(1 - 0.5) = 2.125 = y(0.5)$

$z_1 = 1 + \frac{0.5}{2}(-4 - 5.75) = -1.4375 = z(0.5)$

For $n = 1$;

$y_2 = y_1 + \frac{h}{2}(m_1 + m_2)$ ——— (3)

$z_2 = z_1 + \frac{h}{2}(k_1 + k_2)$ ——— (4)

where,

$m_1 = f_1(x_1, y_1, z_1) = f_1(0.5, 2.125, -1.4375) = -0.9375$

$k_1 = f_2(x_1, y_1, z_1) = f_2(0.5, 2.125, -1.4375) = -4.0156$

$m_2 = f_1(x_1 + h, y_1 + m_1 h, z_1 + k_1 h) = f_1(1, 1.6563, -3.4453)$

$= -2.4453$

$k_2 = f_2(x_1 + h, y_1 + m_1 h, z_1 + k_1 h) = f_2(1, 1.6563, -3.4453)$

$= -1.7433$

From (3) & (4), we get:-

$y_2 = 2.125 + \frac{0.5}{2}(-0.9375 + -2.4453) = 1.2793 = y(1)$. Ans

$z_2 = -1.4375 + \frac{0.5}{2}(-4.0156 - 1.7433) = -2.8872 = z(1)$

Ans

Second order differential equation:

⇒ Let the second order differential equation be

$$\frac{d^2y}{dx^2} = f_2(x, y, y')$$

such that $y(x_0) = y_0$, $y'(x_0) = y_0'$

Put $y' = z = f_1(x, y, z)$

$\therefore z' = f_2(x, y, z)$

$y(x_0) = x_0$; $z(x_0) = z_0$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{dz}{dx}$$

* Numericals:

2023-Spring:

Q.1) Using RK 4th order method to solve the equation:

$\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} - 3y = 6x$ with $y(0) = 0$, $y'(0) = 1$, $h = 0.1$. Estimate $y(0.1)$

Sol Here,

$$\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} - 3y = 6x$$

$$\Rightarrow \frac{d^2y}{dx^2} = 2x\frac{dy}{dx} + 3y + 6x = f_2(x, y, y')$$

Put $\frac{dy}{dx} = y' = z = f_1(x, y, z)$

$\therefore z' = f_2(x, y, z) = 2xz + 3y + 6x$

$\therefore y_0 = 0$; $x_0 = 0$; $y_0' = 1$; $h = 0.1$

By RK 4th order method,

$$y_1 = y_0 + \frac{h}{6}(m_1 + 2m_2 + 2m_3 + m_4) \quad \text{——— (1)}$$

$$\therefore z_1 = z_0 + \frac{h}{6}(K_1 + 2K_2 + 2K_3 + K_4) \quad \text{——— (2)}$$

where,

$m_1 = f_1(x_0, y_0, z_0) = f_1(0,0,1) = 1 = z$

$k_1 = f_2(x_0, y_0, z_0) = f_2(0,0,1) = 0 = z'$

$m_2 = f_1(x_0+\frac{h}{2}, y_0+\frac{m_1 h}{2}, z_0+\frac{k_1 h}{2}) = f(0.05, 0.05, 1)$
$= 1$

$k_2 = f_2(x_0+\frac{h}{2}, y_0+\frac{m_1 h}{2}, z_0+\frac{k_1 h}{2}) = f(0.05, 0.05, 1)$
$= 0.55$

$m_3 = f_1(x_0+\frac{h}{2}, y_0+\frac{m_2 h}{2}, z_0+\frac{k_2 h}{2}) = f_1(0.05, 0.05, 1.0275)$
$= 1.0275$

$k_3 = f_2(x_0+\frac{h}{2}, y_0+\frac{m_2 h}{2}, z_0+\frac{k_2 h}{2}) = f_2(0.05, 0.05, 1.0275)$
$= 0.5528$

$m_4 = f_1(x_0+h, y_0+m_3 h, z_0+k_3 h) = f_1(0.1, 0.1028, 1.0553)$
$= 1.0553$

$k_4 = f_2(x_0+h, y_0+m_3 h, z_0+k_3 h) = f_2(0.1, 0.1028, 1.0553)$
$= 1.1195$

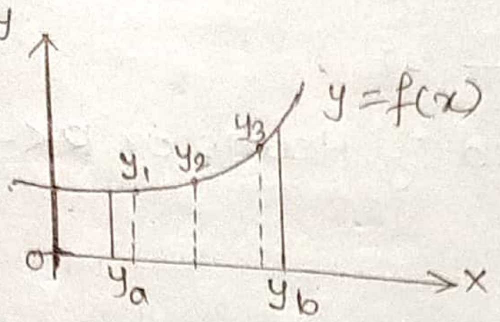
From (1),

$y_1 = 0 + \frac{0.1}{6}(1 + 2\times 1 + 2\times 1.0275 + 1.0553)$
$= 0.1018 = y(0.1)$ Ans

$z_1 = 1 + \frac{0.1}{6}(0 + 2\times 0.55 + 2\times 0.5528 + 1.1195)$
$= 1.0554 = y'(0.1)$ Ans

Boundary Value Problems: Shooting method:

⇒ A second order differential equation of the form $y'' = f(x, y, y')$, $y(a) = A$, $y(b) = B$ is called boundary value Problems.



shooting Method:

→ shooting method is used for solving the boundary value problem.

→ In this method, the given boundary value problem is first converted into an equivalent initial value problem then solved.

→ Consider the equation

$$y'' = f(x, y, y')$$

$$y(a) = A$$

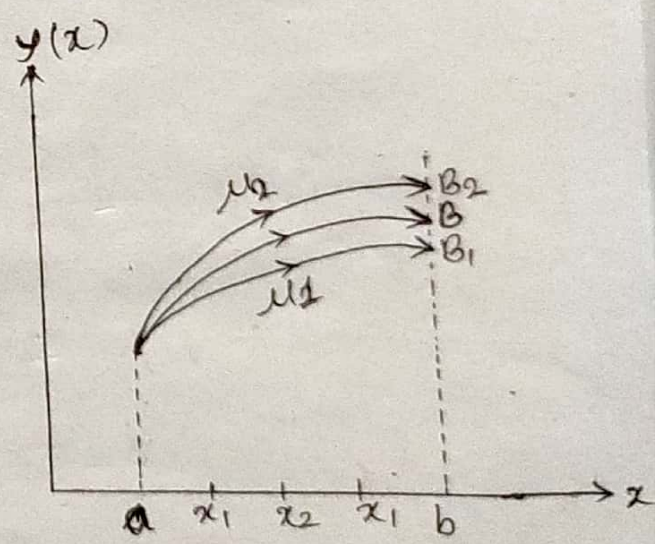
& $y(b) = B$

$$y' = z$$

Then,

$$y' = z$$

$$z' = f(x, y, z)$$



(Fig. Illustration of shooting method.)

In order to solve this, set as an initial value problem. We need two test conditions at $x = a$ and therefore, required another condition for 'z' at $x = a$.

Let us assume that $z(a) = \mu_1$, where μ_1 is a queue.

Solution of Partial Differential Equations

⇒ The general form of second order partial differential equation is:

$$aU_{xx} + bU_{xy} + cU_{yy} = f(x, y, u_x, u_y)$$

Then the equation is said to be

i> Elliptical if $b^2 - 4ac < 0$

ii> Parabolic if $b^2 - 4ac = 0$

iii> Hyperbolic if $b^2 - 4ac > 0$

→ Eg: Laplace equation; $u_{xx} + u_{yy} = 0$

Poisson's equation: $u_{xx} + u_{yy} = f(x, y)$

* Solution of partial differential equation by finite difference method:

⇒ $$\frac{\partial u}{\partial x} = \frac{u(x+h, y) - u(x-h, y)}{2h} \quad \left(\because f'(x) = \frac{f(x+h) - f(x-h)}{2h}\right)$$

$$\therefore \frac{\partial u}{\partial x} = \frac{u(i+1, j) - u(i-1, j)}{2h}$$

$$\therefore \frac{\partial^2 u}{\partial x^2} = \frac{u(i+1, j) + u(i-1, j) - 2u(i, j)}{h^2} \quad ; h = \text{step size}$$

Similarly,

$$\therefore \frac{\partial^2 u}{\partial y^2} = \frac{u(i, j+1) + u(i, j-1) - 2u(i, j)}{h^2} \quad ; k = \text{step size}$$

For square mesh, $h = k$

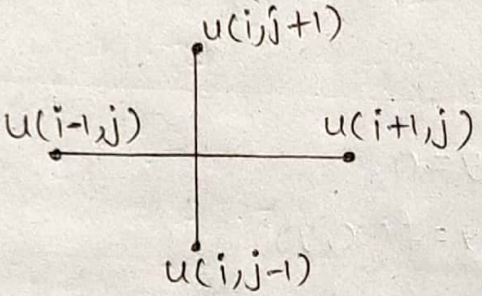
1) From Laplace Equation:

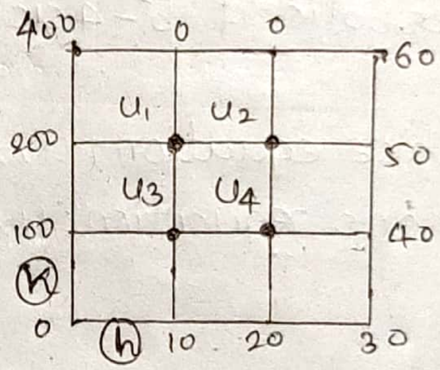
$$u_{xx} + u_{yy} = 0$$

$$\Rightarrow \frac{1}{h^2}\left[u(i+1,j) + u(i-1,j) + u(i,j+1) + u(i,j-1) - 4u(i,j)\right] = 0$$

$$\Rightarrow \therefore u(i+1,j) + u(i-1,j) + u(i,j+1) + u(i,j-1) = 4u(i,j)$$

$$\therefore u(i,j) = \frac{1}{4}\left[u(i+1,j) + u(i-1,j) + u(i,j+1) + u(i,j-1)\right]$$





2) From Poisson's Equation:

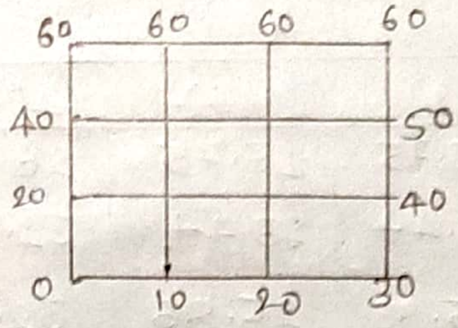
$$u_{xx} + u_{yy} = f(x,y)$$

It's solution is given by

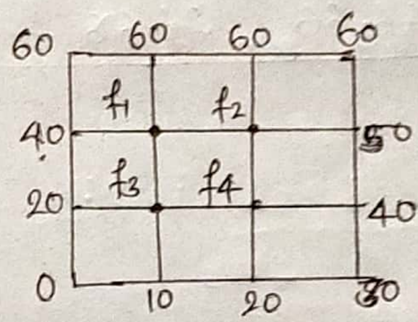
$$u(i+1,j) + u(i-1,j) + u(i,j+1) + u(i,j-1) - 4u(i,j) = h^2 f(ih, jh)$$

* Numericals:

Q.1) Solve $\nabla^2 f = 0$ i.e. $f_{xx} + f_{yy} = 0$ for the square domain shown in the figure given below:



sol: Let f_1, f_2, f_3 and f_4 be the interior points of the given domain.



1) Using Laplace Equation's solution:

$$f_1 = \frac{1}{4}(f_2 + 40 + 60 + f_3)$$

$$\therefore f_1 = \frac{1}{4}(f_2 + f_3 + 100) \quad \text{——— (1)}$$

$$f_2 = \frac{1}{4}(50 + f_1 + 60 + f_4)$$

$$\therefore f_2 = \frac{1}{4}(f_1 + f_4 + 110) \quad \text{——— (2)}$$

$$f_3 = \frac{1}{4}(f_4 + 20 + f_1 + 10)$$

$$\therefore f_3 = \frac{1}{4}(f_1 + f_4 + 30) \quad \text{——— (3)}$$

$$f_4 = \frac{1}{4}(40 + f_3 + \cancel{60}\, f_2 + 20)$$

$$\therefore f_4 = \frac{1}{4}(f_2 + f_3 + 60) \quad \text{——— (4)}$$

Solving these equations by Gauss-Seidel Method;
Let $f_2 = f_3 = f_4 = 0$ be the initial guess.

Iteration Table:

It no.	$f_1(1)$	$f_2(2)$	$f_3(3)$	$f_4(4)$
1.	25	33.75	13.75	26.88
2.	36.88	43.44	23.44	31.72
3.	41.72	45.88	25.86	32.93
4.	42.93	46.47	26.47	33.24
5.	43.24	46.62	26.62	33.31
6.	43.31	46.66	26.66	33.33

Hence,

$f_1 \approx 43.3$

$f_2 \approx 46.6$

$f_3 \approx \cancel{26}.26.6.$

$f_4 \approx 33.3$

These are the required solution of given differential equation.

2019-Spring

Q.2.> Solve the poisson's equation $u_{xx}+u_{yy}=243(x^2+y^2)$ over a square domain $0\le x\le 1$, $0\le y\le 1$ with step size $h=1/3$ with $u=100$ on the boundary.

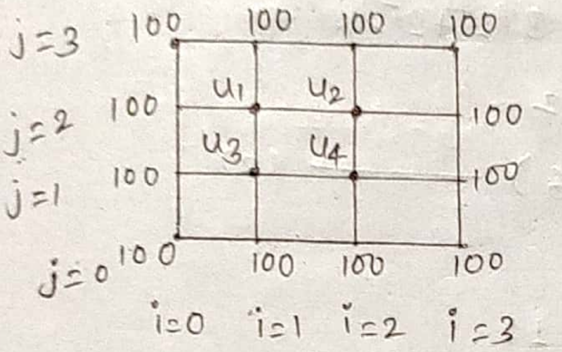
Sol^n

Here,

$$u_{xx}+u_{yy}=243(x^2+y^2);\ 0\le x\le 1;\ 0\le y\le 1$$

$$h=1/3;\ u=100$$

Let u_1, u_2, u_3 & u_4 be the interior points of the given domain. The square domain will be:



Note i&j ko values always start from 0.

The solution of Poisson's equation is given by

$$u(i+1,j)+u(i-1,j)+u(i,j+1)+u(i,j-1)-4u(i,j)=h^2f(ih,jh)$$

$$=(1/3)^2\cdot 243\{(ih)^2+(jh)^2\}$$

$$=\frac{243}{9}\cdot h^2(i^2+j^2)$$

$$=\frac{243}{9}\times\frac{1}{9}(i^2+j^2)$$

$$=3(i^2+j^2)$$

$$\therefore u(i+1,j)+u(i-1,j)+u(i,j+1)+u(i,j-1)-4u(i,j)$$

$$=3(i^2+j^2) \quad \text{———} \quad (A)$$

~~Let u_1, u_2, u_3 and u_4 be the interior points of the given domain~~

~~Let u~~1,

i> Take $u(i,j) = u_1 ; i=1 ; j=2$:

From (A),

$$u_2+100+100+u_3-4u_1 = 3(1^2+2^2)$$

$$\Rightarrow u_2+u_3+200-3\times 5 = 4u_1$$

$$\therefore u_1 = \frac{1}{4}(u_2+u_3+185) \quad \text{———} \quad (1)$$

ii> Take $u(i,j) = u_2 ; i=2 ; j=2$:

From (A),

$$100+u_1+100+u_4-4u_2 = 3(2^2+2^2)$$

$$\Rightarrow u_1+u_4+200-3\times 8 = 4u_2$$

$$\therefore u_2 = \frac{1}{4}(u_1+u_4+176) \quad \text{———} \quad (2)$$

iii> Take $u(i,j) = u_3 ; i=1 ; j=1$:

From (A),

$$u_4+100+u_1+100-4u_3 = 3(1^2+1^2)$$

$$\Rightarrow u_1+u_4+200-3\times 2 = 4u_3$$

$$\therefore u_3 = \frac{1}{4}(u_1+u_4+194) \quad \text{———} \quad (3)$$

iii> Take $u(i,j) = u_4 ; i=2 ; j=1$:

From (A),

$$100+u_3+u_2+100-4u_4 = 3(2^2+1^2)$$

$$\Rightarrow u_2+u_3+200-3\times 5 = 4u_4$$

$$\therefore u_4 = \frac{1}{4}(u_2+u_3+185) \quad \text{———} \quad (4)$$

Solving these equations by Gauss-Seidel Method.

Since $u_1 = u_4$, so

Let $u_2 = u_3 = 0$ be the initial guess.

It no.	$u_1(1)$	$u_2(2)$	$u_3(3)$	$u_4(4)$
1.	46.3	67.2	71.7	46.3
2.	81.0	84.5	89.0	81.0
3.	89.6	88.8	93.3	89.6
4.	91.8	89.9	94.4	91.8
5.	92.3	90.2	94.7	92.3
6.	92.5	90.3	94.8	92.5

on solving, The solutions of given differential equations are:

$u_1 \approx 92.5 = u_4$

$u_2 \approx 90.3$

$u_3 \approx 94.8$

Q.3.> Solve the Poisson's equation $\nabla^2 u = -81xy$; $0 \le x \le 1$, $0 \le y \le 1$, $u(0,y) = 0$, $u(x,0) = 0$, $u(1,y) = 100$ & $u(x,1) = 100$ such that dividing the given square domain into nine equal square.

sol Here,

$$\nabla^2 u = -81xy; \ 0 \le x \le 1; \ 0 \le y \le 1$$

$u(0,y) = 0$, $u(x,0) = 0$

$u(1,y) = 100$, $u(x,1) = 100$

To find h: Divide $(1-0)$ into 3 equal parts:

$\therefore h = 1/3$

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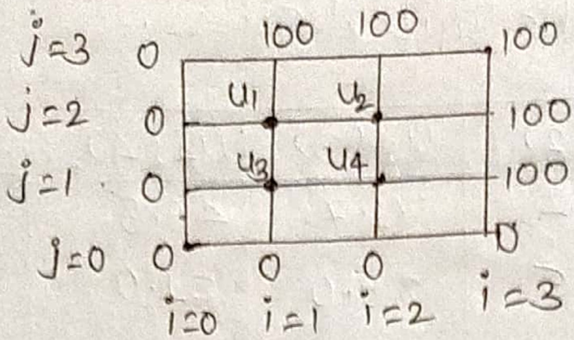
Let u_1, u_2, u_3 and u_4 be the interior points of the given domain as shown in figure below:



The solution of poisson's equation is given by

$$u(i+1,j)+u(i-1,j)+u(i,j+1)+u(i,j-1)-4u(i,j)$$
$$= h^2 f(ih, jh)$$
$$= h^2(-81 (ih).(jh))$$
$$= h^4(-81 ij)$$
$$= \left(\frac{1}{3}\right)^4 . -81 ij$$
$$= -ij$$

$\therefore$

$$\therefore u(i+1,j)+u(i-1,j)+u(i,j+1)+u(i,j-1)-4u(i,j)$$
$$= -ij \quad \text{———} \quad (A)$$

i) Take $u(i,j) = u_1$; $i=1$; $j=2$:

From (A),

$$u_2 + 0 + 100 + u_3 - 4u_1 = -1 \times 2$$
$$\Rightarrow u_2 + u_3 + 100 + 2 = 4u_1$$
$$\therefore u_1 = \frac{1}{4}(u_2 + u_3 + 102) \quad \text{———} \quad (1)$$

ii) Take $u(i,j) = u_2$; $i=2$; $j=2$:

from (A),

$100 + u_1 + 100 + u_4 - 4u_2 = -2\times 2$

$\Rightarrow u_1 + u_4 + 200 + 4 = 4u_2$

$\therefore u_2 = \frac{1}{4}(u_1 + u_4 + 204)$ ——— (2)

iii) Take $u(i,j) = u_3$; $i=1$; $j=1$:

from (A),

$u_4 + 0 + u_1 + 0 - 4u_3 = -1\times 1$

$\Rightarrow u_1 + u_4 + 1 = 4u_3$

$\therefore u_3 = \frac{1}{4}(u_1 + u_4 + 1)$ ——— (3)

iii) Take $u(i,j) = u_4$; $i=2$; $j=1$:

from (A),

$100 + u_3 + u_2 + 0 - 4u_4 = -2\times 1$

$\Rightarrow u_2 + u_3 + 100 + 2 = 4u_4$

$\therefore u_4 = \frac{1}{4}(u_2 + u_3 + 102)$ ——— (4)

Since $u_1 = u_4$.

solving these equations (1), (2) & (3) & (4) by Gauss-Seidel Method.

Let $u_2 = u_3 = 0$ be the initial guess.

It no.	$u_1(1)$	$u_2(2)$	$u_3(3)$	$u_4(4)$
1.	25.5	63.8	13	25.5
2.	44.7	73.4	22.6	44.7
3.	49.5	75.8	25	49.5
4.	50.7	76.4	25.6	50.7
5.	51	76.5	25.8	51
6.	51.1	76.6	25.8	51.1
7.	51.1	76.6	25.8	51.1

on solving,

$$u_1 = 51.1 = u_4$$

$$u_2 = 76.6$$

$$u_3 = 25.8$$

These are the required solution of given differential equations. Ans

Solution of Non-Linear Equation

: 10 hrs

Error Analysis:

Error = |Exact Value − Approximate Value|

Types of Numerical Error:

1> Truncation Error:

→ The word 'Truncate' means 'to shorten'.

→ Truncation error refers to an error in a method which occurs because some number/series of steps (finite or infinite) is truncated (shortened) to a fewer number.

→ Such errors are essentially algorithmic errors and we can predict the extent of the error that will occur in the method.

→ For example, if we approximate the exponential function by the first four non-zero term of its Maclaurin series as

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$$

$$\therefore \text{Truncation Error} = e^x - \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!}\right)$$

$$= \left(\frac{x^5}{5!} + \frac{x^6}{6!} + \cdots\right)$$

→ The resulting error is a truncation error. It is present even with infinite-precision arithmetic, because it is caused by truncation of the infinite Taylor series to form the algorithm.

2> Round-off Error:

→ Round-off error occurs because of the computing device's ability to deal with certain numbers.

$1/3 = 0.333333.....$

$1/6 = 0.166666-...$

$\pi(\text{Pi}) = 3.14159265358932 38.......$

→ Such numbers need to be rounded-off to some near approximation which is dependent on the word size used to represent numbers of the device.

→ When the number of bits required for representing a number are less than the number is usually rounded to fit the available number of bits.

→ This is done ~~by~~ either by chopping or by symmetric rounding.

i> Chopping Error:

→ In chopping error, the digits that are beyond the storage capacity of the computer are dropped.

→ If the word length of the computer is 4 digits, then a number like 32.5472 will be stored as 32.54. Digits 7 and 2 will be dropped.

→ Chopping error = $32.5472 - 32.54 = 0.0072$

ii> Symmetric Round-off Error:

→ In a symmetric round-off error, the last retained significant digit is rounded by 1, if the first digit being discarded is greater than or equal to 5 and the last retained digit is unchanged, if the first digit being discarded is less than 5.

→ In the previous example, 32.5472 will become 32.55 because 7 is greater than 5.

→ If the original number was 32.5432, then it will be stored as 32.54.

→ Round-off error (symmetric) $= |32.5472 - 32.55|$
$= 0.0028$

Absolute Error and Relative Error:

→ Let x_t be the true value and x_a be the approximate value of any problem. Then,

i> Absolute Error, $E_a = |x_t - x_a|$

ii> Relative Error, $E_r = \left|\dfrac{x_t - x_a}{x_t}\right| \times 100\%$

Q. An approximate value of π is given by 3.14285 and the true value is 3.1415926. Find the absolute and relative error. [P.U. BCA]

Sol Given;
Approximate Value, $x_a = 3.142857$
True value, $x_t = 3.1415926$

$\therefore$ Absolute Error, $E_a = |x_t - x_a|$
$= |3.1415926 - 3.142857|$
$\therefore E_a = 0.0012644$

$\therefore$ Relative Error, $E_r = \dfrac{E_a}{x_t} \times 100\% = \dfrac{0.0012644}{3.1415926} \times 100\%$
$\therefore E_r = 0.040247027\%$